Straight lines in the plane

Lecture and worked examples

Direction vector and Cartesian equation

A direction vector of a line is a **nonzero** vector having the direction of that line. If $A \neq B$, $\overrightarrow{AB}$ directs (AB). Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

Direction vector and Cartesian equation

For $A(-2,4)$ and $B(6,2)$, calculate $\overrightarrow{AB}$ then check that $\vec{u}(4,-1)$ directs (AB).

Worked answer: Direction vector and Cartesian equation

$\overrightarrow{AB} = (8, -2) = 2\vec{u}$. The vector $\vec{u}$ is nonzero and collinear with $\overrightarrow{AB}$; it therefore directs (AB).

Direction vector and Cartesian equation

The line through $A(x_A, y_A)$ with direction vector $\vec{u}(\alpha, \beta) \neq \vec{0}$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

Direction vector and Cartesian equation

A point $M(x, y)$ belongs to the line if and only if $\overrightarrow{AM}$ is collinear with $\vec{u}$, including for $M = A$. Now

$$\det(\overrightarrow{AM}, \vec{u}) = (x - x_A)\beta - (y - y_A)\alpha.$$

Setting the determinant to zero gives the stated equation. Expanding, $a = \beta$, $b = -\alpha$ and $c = -\beta x_A + \alpha y_A$; since $\vec{u} \neq \vec{0}$, $(a, b) \neq (0, 0)$.

Direction vector and Cartesian equation

The zero vector directs no line. The sign of the determinant depends on the order of the vectors, but whether it vanishes does not.

Determining the equation of a line

From a point and a direction vector: choose a, b then replace x, y by the coordinates of the point to find c. From two distinct points: first calculate a direction vector.

Determining the equation of a line

Determine a Cartesian equation of the line through $A(2, -1)$ with direction vector $\vec{u}(3, 2)$.

Worked answer: Determining the equation of a line

$a = 2$, $b = -3$ will do. The line therefore has an equation $2x - 3y + c = 0$. Since A belongs to it, $4 + 3 + c = 0$, so $c = -7$. An equation is $2x - 3y - 7 = 0$. Check: $2 \times 2 - 3 \times (-1) - 7 = 0$ and $(3, 2)$ is indeed a direction vector.

Determining the equation of a line

Determine a Cartesian equation of (AB) for $A(3, 3)$ and $B(2, -2)$.

Worked answer: Determining the equation of a line

$\overrightarrow{AB} = (-1, -5)$ directs the line. We can choose $(a, b) = (5, -1)$. With A, $15 - 3 + c = 0$, so $c = -12$. Thus $5x - y - 12 = 0$. For B, $10 + 2 - 12 = 0$, which confirms the result.

Determining the equation of a line

Does the point $P(1, 2)$ belong to $d : 2x - 3y + 4 = 0$? Does the point $Q(2, 1)$ belong to it?

Worked answer: Determining the equation of a line

For P, $2 - 6 + 4 = 0$: yes. For Q, $4 - 3 + 4 = 5 \neq 0$: no.

Slope-intercept equation, slope and affine function

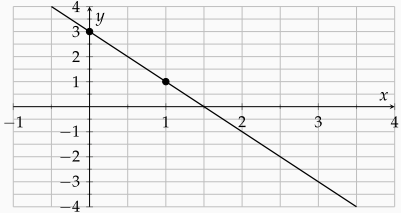
A non-vertical line has a unique slope-intercept equation $y = mx + p$. m is the slope, p the y-intercept. It represents the affine function $x \mapsto mx + p$; $(1, m)$ is a direction vector. A vertical line has an equation $x = k$; it is not the graph of a function of x. A horizontal line has an equation $y = k$.

Slope-intercept equation, slope and affine function

If $x_A \neq x_B$, the slope of (AB) is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find p, we then use $y_A = mx_A + p$.

Slope-intercept equation, slope and affine function

Determine the slope-intercept equation of $2x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.



Worked answer: Slope-intercept equation, slope and affine function

$3y = -2x + 1$, so $y = -\frac{2}{3}x + \frac{1}{3}$. For d, plot $(0, 3)$ and $(1, 1)$, then draw the line; a move of 1 along the x-axis corresponds to -2 on the y-axis.

Slope-intercept equation, slope and affine function

Find the affine function f such that $f(0) = 3$ and $f(-1) = 5$, then the affine function g whose graph passes through $C(-1, 2)$ and $E(5, -2)$.

Worked answer: Slope-intercept equation, slope and affine function

$f(x) = -2x + 3$, since $m = (3-5)/(0+1) = -2$. For g, $m = (-2-2)/(5+1) = -2/3$ and $p = 2 - 2/3 = 4/3$, so $g(x) = -\frac{2}{3}x + \frac{4}{3}$.

Relative positions of two lines

Two lines are either intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$:

- if $m \neq m'$, they intersect;
- if $m = m'$ and $p \neq p'$, they are strictly parallel;
- if $m = m'$ and $p = p'$, they coincide.

For Cartesian equations, we compare the directions with the determinant. If the directions are collinear, we then test the membership of a point to distinguish coincident lines from strictly parallel ones.

Relative positions of two lines

Determine an equation of d through $A(2,3)$ and parallel to $d' : x - 3y - 3 = 0$.

Worked answer: Relative positions of two lines

d' has direction vector $(3, 1)$. We look for $d : x - 3y + c = 0$. Since $A \in d$, $2 - 9 + c = 0$, so $c = 7$. Thus $d : x - 3y + 7 = 0$. The two lines are strictly parallel, because the constants differ for the same coefficients of x, y.

Relative positions of two lines

Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$;
2. $x = 2$ and $x = -1$;
3. $y = 2x - 3$ and $y = -x + 3$.

Worked answer: Relative positions of two lines

1. The second equation is twice the first: coincident lines. 2. Two distinct vertical lines: strictly parallel. 3. The slopes 2 and -1 differ: intersecting. Their intersection satisfies $2x - 3 = -x + 3$, so $x = 2$ and $y = 1$.

Intersection: solving by substitution

Solving a system of two equations in x, y means finding all the pairs (x, y) satisfying both equations simultaneously. For two equations of lines: an intersection gives one solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

Intersection: solving by substitution

Isolate one unknown in one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

Intersection: solving by substitution

Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

Worked answer: Intersection: solving by substitution

The first equation gives $y = 2x - 3$. Substituting into the second: $3x - 5(2x - 3) - 2 = 0$, that is $-7x + 13 = 0$. Thus $x = 13/7$ then $y = 5/7$. The solution set is $S = \{(13/7, 5/7)\}$. The substitutions give $(-26 + 5 + 21)/7 = 0$ and $(39 - 25 - 14)/7 = 0$.

Intersection: solving by substitution

Solve by substitution:

(a) $\begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases}$ (b) $\begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$

Worked answer: Intersection: solving by substitution

(a) $y = 2x - 3$ then $x - 4(2x - 3) + 5 = 0$, so $x = 17/7$, $y = 13/7$.

(b) $x = (10 - 3y)/2$; $2(10 - 3y) + 5y - 16 = 0$, so $y = 4$, then $x = -1$.

Linear combination and Python

Multiply the equations by nonzero numbers to obtain opposite coefficients for one unknown; add them to eliminate that unknown. Keep one equation of the system to calculate the other unknown.

Linear combination and Python

Solve $\begin{cases} 2x+3y+1=0 \\ 3x-4y-2=0. \end{cases}$ Then redo the two systems of the previous example by linear combination.

Worked answer: Linear combination and Python

Multiplying the first equation by 3 and the second by -2 gives $6x + 9y + 3 = 0$ and $-6x + 8y + 4 = 0$. The sum gives $17y + 7 = 0$, so $y = -7/17$. Then $2x - 21/17 + 1 = 0$, so $x = 2/17$. For (a), add four times the first equation to the second: $-7x + 17 = 0$, so $(x, y) = (17/7, 13/7)$. For (b), subtract twice the first from the second: $-y + 4 = 0$, so $(x, y) = (-1, 4)$.

Python

```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

Linear combination and Python

Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

Worked answer: Linear combination and Python

`True`, then `False`: the determinants are 0 and 2. The exact test is suitable here for integer coordinates; floating-point approximations require taking numerical errors into account.